Jake Brawer, Ph.D.

website / github / email / linkedin

RESEARCH OVERVIEW

Summary

I build **interactive AI systems** that combine perception, reasoning, memory, and learning to support robust human-centered interaction. My work spans **robotics**, **conversational AI**, and **production ML systems**, with emphasis on realtime decision-making, structured knowledge, personalization, and deployable architectures.

PROFESSIONAL EXPERIENCE

Founding Engineer

Stealth startup (voice AI / digital health), Nov 2025–Current
Sole architect and founding engineer for an early-stage phone-based conversational AI platform for older adults.
Built the end-to-end system across telephony, realtime speech, LLM orchestration, long-term memory, FastAPI services, PostgreSQL, and AWS infrastructure.
Implemented platform safeguards for reliability and HIPAA-oriented handling of sensitive user data.

Postdoc

University of Colorado, Boulder, 2023–2025
Advisors: Alessandro Roncone and Bradley Hayes
Lead research on human–robot teaming, assistive robotics, and AI decision support.
Develop models for inferring human proficiency, trust, and intent from gaze and behavior in collaborative environments.
Build semantic robot perception systems for robust localization in cluttered indoor workspaces using low-cost visual-inertial sensing.

Lecturer

University of Colorado, Boulder, Spring 2024
Sole lecturer for Advanced Robotics (CSCI 4302/5302), a graduate-level course covering motion planning, SLAM, optimization, and reinforcement learning.

EDUCATION

Yale

Ph.D. in Computer Science, 2016–2023
Advisor: Brian Scassellati

Vassar

B.A. in Cognitive Science; Computer Science minor, 2012–2016

SKILLS

Languages

Python, C/C++, C#, SQL, Git, Emacs, Docker, Linux, LaTeX

AI / ML

PyTorch, scikit-learn, SciPy, NumPy, Pandas, LangChain, LLM orchestration, RAG / vector retrieval, PGVector, matplotlib, NLTK, Gurobi, Prolog, PDDL

Backend

FastAPI, PostgreSQL, AWS, async Python, REST APIs, system design, deployment, load testing, observability, HIPAA-oriented system design, PHI-safe logging

Voice AI

Pipecat, Deepgram, AWS Bedrock, Telnyx, Cartesia

Robotics/CV

ROS, Gazebo, RViz, OpenCV, Open3D, Baxter, Sawyer, UR5, Kinect, RealSense, RGB-D cameras

Soft Skills

Clear technical communication, Collaborative, Fast learner, Leadership

Selected publications

Shivendra Agrawal, **Jake Brawer**, Ashutosh Naik, Alessandro Roncone, and Bradley Hayes (2026). "**ShelfAware: Real-Time Visual-Inertial Semantic Localization in Quasi-Static Environments with Low-Cost Sensors**". In: *IEEE Robotics and Automation Letters*. Accepted.

Nikhil Hulle, Stéphane Aroca-Ouellette, Anthony J Ries, **Jake Brawer**, Katharina von der Wense, and Alessandro Roncone (2024). "**Eyes on the Game: Deciphering Implicit Human Signals to Infer Human Proficiency, Trust, and Intent**". In: *33rd IEEE International Conference on Robot and Human Interactive Communication (RO-MAN 2024)*. Accepted for publication, proceedings pending.

Jake Brawer, Debasmita Ghose, Kate Candon, Meiying Qin, Alessandro Roncone, Marynel Vazquez, and Brian Scassellati (2023). "**Interactive Policy Shaping for Human-Robot Collaboration with Transparent Matrix Overlays**". In: *Proceedings of the 2023 ACM/IEEE International Conference on Human-Robot Interaction*. IEEE.

Jake Brawer, Meiying Qin, and Brian Scassellati (2020). "**A causal approach to tool affordance learning**". In: *2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, pp. 8394–8399.

AWARDS AND SERVICE

2025

Outstanding Reviewer Award, Robotics: Science and Systems (**RSS**)

2023-2025

Army Educational Outreach Program (AEOP) Fellowship, University of Colorado Boulder

2024

Area Chair, International Conference on Human–Agent Interaction (**HAI**)

2023

Best paper (technical track), ACM/IEEE International Conference on Human-Robot Interaction (**HRI**) for “Interactive Policy Shaping for Human-Robot Collaboration”

Workshops

Co-organized the Human-Machine Teaming Paradigm Workshop (2023) and Bridging the Gap: An NSF Workshop on Conversational Agents and Human-Robot Interaction (2018).

Service

Reviewer for **RSS**, **HRI**, **IROS**, **ICRA**, **Humanoids**, **IJRR**, **T-RO**, and **THRI**; mentored undergraduate and graduate researchers on robotics, HRI, and assistive AI projects.